INTEGRATIVE PHYSIOLOGY (POST LEC)

Lecture #3: RESPIRATORY VENTILATION | OPHT 1-1 | MIDTERMS

By Accel Jimenez

THE FOUR COMPONENTS OF REPIRATION

Pulmonary Ventilation: The mechanical process of moving air in and out of the lungs (breathing).
Gas Diffusion: The exchange of oxygen (O2) and carbon dioxide (CO2) between the lung alveoli and the blood.
Gas Transport: The circulation of O2 and CO2 via the blood to and from the body's tissues.
Regulation: The control of breathing and other aspects of respiration.

Two types of Respiration
External – between the alveolar and capillaries
Internal – between the capillaries and tissues

Mechanics of Breathing: Muscles

- Breathing involves changing the volume of the chest cavity, which is achieved in two main ways:

Movement of Diaphragm:

- This is the primary method for **normal, quiet breathing.**
- **Inspiration (Inhalation):** The diaphragm contracts and moves downward, increasing the vertical length of the chest cavity and pulling air into the lungs.
- **Expiration (Exhalation):** In **quiet breathing**, this is a passive process. The diaphragm relaxes, and the elastic recoil of the lungs and abdominal structures pushes air out.

Movement of the Rib Cage:

- **Muscles of Expiration** (depress the rib cage):
    - Used during heavy or forced breathing when passive recoil is not enough.
    - **Abdominal Muscles** (e.g., abdominal recti): These contract to push the abdominal organs up against the diaphragm, forcing air out.
    - **Internal Intercostals:** These pull the rib cage downward.

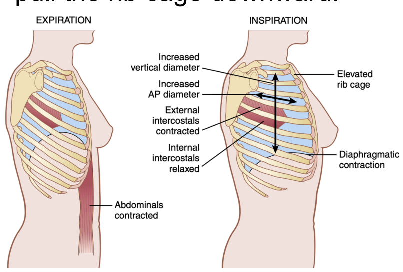

Figure 38-1 Contraction and expansion of the thoracic cage during expiration and inspiration, demonstrating diaphragmatic contraction, function of the intercostal muscles, and elevation and depression of the rib cage. AP, Anteroposterior.

Mechanics of Breathing:Pressures

- Air moves in and out of the lungs based on pressure differences relative to the atmosphere.

Pleural Pressure:

- This is the pressure of the fluid in the space between the lungs and the chest wall.
- It is always slightly negative (a suction), which holds the lungs against the chest wall.
- **During inspiration**, the chest expands, making the **pleural pressure more negative** (e.g., from -5 to -7.5 cm H2O).

Alveolar Pressure:

- This is the air pressure inside the lung alveoli.
- **To inspire,** alveolar pressure must become negative (lower than atmospheric pressure, e.g., -1 cm H2O), causing air to flow in.
- **To expire,** alveolar pressure must become positive (higher than atmospheric pressure, e.g., +1 cm H2O), forcing air to flow out.

Transpulmonary Pressure:

- This is the difference between the alveolar pressure and the pleural pressure.
- It's **a measure of the elastic forces** (recoil pressure) that tend to collapse the lungs.

Lung Compliance: The Basics

- Definition: Lung compliance is a measure of how much the lungs will expand for a given change in transpulmonary pressure. It's essentially the **"stretchiness" of the lungs.**
- **Normal Value:** For a healthy adult, the **average lung compliance is about 200 ml of air per 1 cm H2O of transpulmonary pressure.**
- **Compliance Diagram:** This graph (Figure 38-3) shows the relationship between lung volume and pressure. The path for inspiration is different from the path for expiration, a phenomenon known as hysteresis.

The Forces That Determine Lung Compliance

- The elasticity of the lungs, which determines their compliance, comes from two main sources:

1. **Tissue Elastic Forces** (≈ 1/3 of total elasticity):

- Caused by the elastin and collagen fibers within the lung tissue.
- These fibers stretch during inhalation and recoil during exhalation.

2. **Surface Tension Forces** (≈ 2/3 of total elasticity):
   - This is the most significant factor affecting lung compliance.
   - It's caused by the attraction between water molecules lining the inside of the alveoli. This force creates a "tension" that constantly tries to collapse the alveoli.
3. **Surface Tension Forces** (≈ 2/3 of total elasticity):
   - The experiment in Figure 38-4 demonstrates this:
     - Air-filled lungs have an air-fluid interface, creating high surface tension.
     - Saline-filled lungs have no air-fluid interface, so surface tension is eliminated.
     - Conclusion: It takes about three times more pressure to expand air-filled lungs than saline-filled lungs, proving that surface tension accounts for about two-thirds of the lung's elastic recoil.

The Critical Role of Surfactant

- What is it? Surfactant is a substance that acts like a detergent to greatly reduce the surface tension of the fluid lining the alveoli.
- Where does it come from? It is secreted by **Type II alveolar epithelial cells.**
- Why is it important? By reducing surface tension, surfactant:
  - Increases lung compliance, making it much easier to inflate the lungs.
  - Prevents the alveoli from collapsing at the end of expiration.

Compliance of the Thorax and Lungs Combined

- The thoracic cage (chest wall) also has its own elasticity and resists expansion.
- When measured together, the compliance of the combined lung-thorax system is much lower than that of the lungs alone.
- **Key Value:** The compliance of the combined system is about **110 ml/cm H2O,** which is almost half the compliance of the lungs alone (200 ml/cm H2O). This means it takes significantly more effort to expand both the lungs and the chest wall together.

Measure Lung Volumes: Spirometry

- Spirometry is the method used to record the volume of air moving in and out of the lungs.
- The device used is a spirometer (Figure 38-5), and the resulting graph of lung volume over time is called a spirogram (Figure 38-6).

The Four Basic Pulmonary Volumes

- These are the fundamental measurements and do not overlap.
- **Tidal Volume (VT):** The amount of air inhaled or exhaled during a normal, quiet breath.
  - Average Value (Men): 500 ml
- **Inspiratory Reserve Volume (IRV):** The extra volume of air that can be forcibly inhaled beyond a normal tidal inspiration.
  - Average Value (Men): 3000 ml
- **Expiratory Reserve Volume (ERV)**: The maximum extra volume of air that can be forcibly exhaled after a normal tidal expiration.
  - Average Value (Men): 1100 ml
- **Residual Volume (RV):** The volume of air that always remains in the lungs, even after the most forceful expiration.
  - Key Concept: This volume cannot be measured by simple spirometry because it cannot be exhaled.
  - Average Value (Men): 1200 ml

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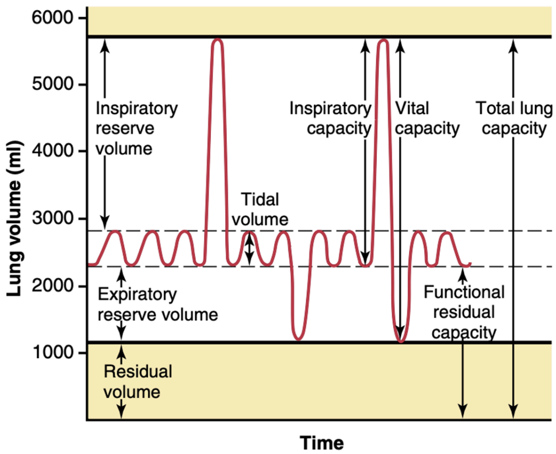

Figure 38-6 Respiratory excursions during normal breathing and during maximal inspiration and maximal expiration.

Table 38-1 Average Pulmonary Volumes and Capacities for Healthy, Young Adult Men and Women

Pulmonary Volumes and Capacities	Men	Women
Volume (ml)		
Tidal volume	500	400
Inspiratory reserve volume	3000	1900
Expiratory volume	1100	700
Residual volume	1200	1100
Capacities (ml)		
Inspiratory capacity	3500	2400
Functional residual capacity	2300	1800
Vital capacity	4600	3100
Total lung capacity	5800	4200

The Four Pulmonary Capacities

- **Capacities** are combinations of two or more volumes and represent a larger functional measurement.
- **Inspiratory Capacity (IC):** The total amount of air a person can inhale, starting from the normal expiratory level.
  - **Formula: IC=VT+IRV**
  - Average Value (Men): 3500 ml
- **Functional Residual Capacity (FRC):** The amount of air remaining in the lungs at the end of a normal, quiet expiration. This is the resting volume of the lungs.
  - **Formula: FRC=ERV+RV**
  - Average Value (Men): 2300 ml
- **Vital Capacity (VC):** The maximum amount of air a person can expel from the lungs after first filling them to their maximum extent. It represents the total usable volume of the lungs.
  - **Formula: VC=IRV+VT+ERV**
  - Average Value (Men): 4600 ml
- **Total Lung Capacity (TLC):** The maximum volume to which the lungs can be expanded with the greatest possible effort. It is the sum of all four volumes.
  - Formula: TLC=VC+RV
  - Average Value (Men): 5800 ml

General Considerations

- All pulmonary volumes and capacities are generally 20-30% less in women than in men.
- Values are typically greater in large and athletic individuals compared to small and non-athletic people.

Minute Respiratory Volume

- Definition: The total volume of air moved into and out of the respiratory passages each minute.
- Formula: Minute Respiratory Volume = Tidal Volume (VT) × Respiratory Rate (Freq)
- **Typical Value:** For a normal adult, this averages about 6 L/min (e.g., 500 ml/breath × 12 breaths/min).
- **Range:** This can vary dramatically, from as low as 1.5 L/min at rest to over 200 L/min during brief, intense exercise.

Alveolar Ventilation & The Concept of Dead Space

- The Goal of Breathing: The ultimate purpose of ventilation is to renew the air in the gas exchange areas of the lungs (the alveoli). The rate at which this happens is called alveolar ventilation.
- **Anatomic Dead Space (VD):**
  - Definition: Not all the air you inhale reaches the alveoli. A portion of it fills the conducting airways (nose, pharynx, trachea, bronchi) where no gas exchange occurs. This volume is called the dead space.
  - **Key Concept:** This air is considered "wasted" ventilation because it does not participate in gas exchange.
  - **Typical Value:** A normal dead space volume is about **150 ml.**

Rate of Alveolar Ventilation (VA):

- **Definition**: This is the most important physiological measure of ventilation because it represents the total volume of new air that actually reaches the alveoli and is available for gas exchange each minute.
- **Formula:** It accounts for dead space by subtracting it from the tidal volume.
  - VA=Respiratory Rate×(Tidal Volume−Dead Space Volume)
  - VA=Freq×(VT−VD)
- **Typical Calculation:** Using normal values, alveolar ventilation is 4200 ml/min (or 4.2 L/min).
  - 12 breaths/min × (500 ml - 150 ml) = 4200 ml/min
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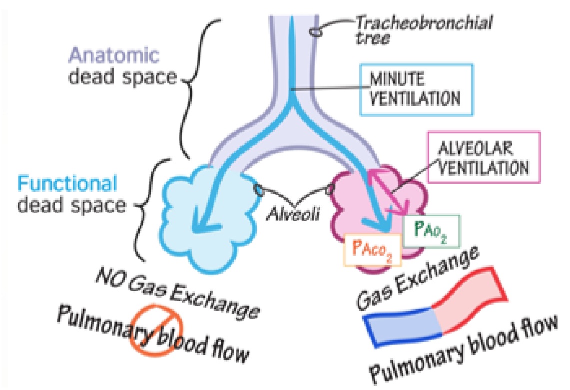
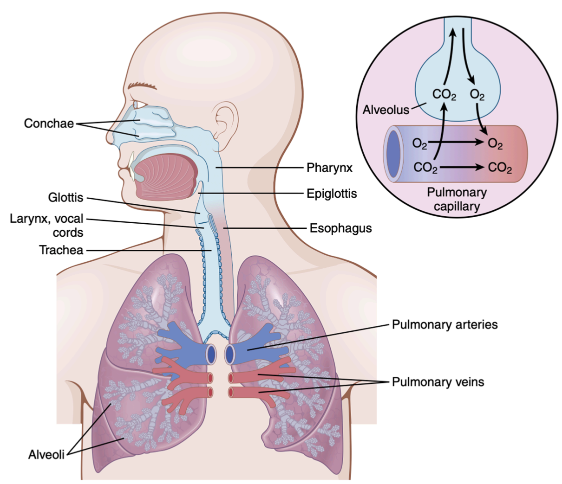




Structure and Airflow Resistance

Structural Support:

- The trachea and bronchi are kept open by rigid cartilage rings and plates.
- The smaller bronchioles lack cartilage and are kept open by the same transpulmonary pressure that expands the alveoli.

Airflow Resistance:

- **Key Concept:** Under normal conditions, the greatest resistance to airflow is in the larger bronchi and bronchioles, not the smallest ones. This is because there are relatively few large airways compared to the massive number (≈65,000) of parallel terminal bronchioles.
- In disease states **(like asthma),** the smaller bronchioles become the major site of resistance due to muscle contraction, edema, or mucus.

Control of Bronchial Smooth Muscle

- The diameter of the bronchioles is actively controlled to regulate airflow.

Dilation (Widening of Airways):

- **Sympathetic Stimulation:** Primarily caused by **epinephrine** and **norepinephrine** circulating in the blood from the adrenal glands. This is a **powerful bronchodilator effect**, acting on beta-adrenergic receptors.

Constriction (Narrowing of Airways):

- **Parasympathetic Stimulation:** Nerve fibers from the vagus nerve release **acetylcholine,** causing mild to moderate **bronchoconstriction.** This can be triggered by irritants like dust and smoke.
- **Local Factors:** Substances like histamine and slow reactive substance of anaphylaxis, released by mast cells during allergic reactions (e.g., asthma), are potent bronchoconstrictors.

Protective Reflexes and Mechanisms

- The respiratory system has several ways to protect itself from foreign material.

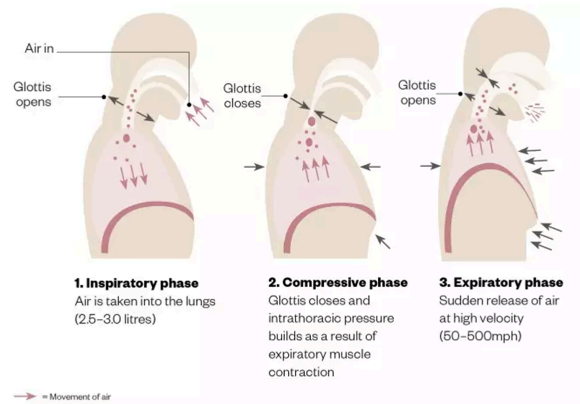
Mucociliary Escalator:

- The airways are lined with a layer of mucus that traps inhaled particles (dust, pollen).
- Tiny hairs called cilia beat continuously in an upward direction, moving the mucus toward the pharynx where it can be swallowed or coughed out.

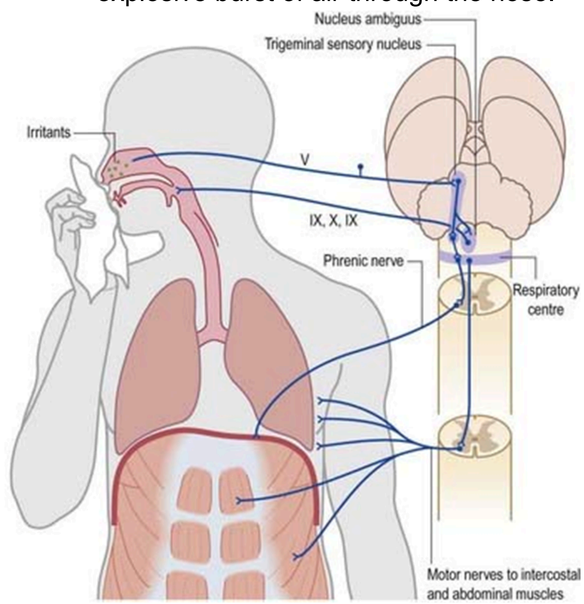
Cough Reflex:

- A protective reflex for the lower respiratory passages (trachea, bronchi).
- **Triggered by: Irritation from foreign matter.**

- Mechanism: A deep breath in, followed by the closure of the epiglottis and forceful contraction of expiratory muscles, building high pressure. The sudden opening creates an explosive outflow of air (up to 75-100 mph) to clear the passages.



Sneeze Reflex:
- Similar to a cough, but it clears the nasal passageways.
- The uvula is depressed to direct the explosive burst of air through the nose.



Blood Flow through the Lungs and Its distribution

Matching Blood Flow to Airflow
- Pulmonary blood flow is essentially equal to the entire cardiac output.
- The primary goal of the pulmonary circulation is to direct blood to the parts of the lungs that are receiving the most oxygen. This matching of blood flow (perfusion) to airflow (ventilation) is critical for efficient gas exchange.

Hypoxic Pulmonary Vasoconstriction: The Key Mechanism
- **The Main Concept:** When the oxygen concentration in a specific area of the alveoli becomes low (hypoxia), the adjacent blood vessels constrict (vasoconstriction).
- **Opposite Effect:** This is the exact opposite of what happens in the rest of the body (systemic circulation), where low oxygen causes blood vessels to dilate.
- **The Purpose (V/Q Matching):** This unique mechanism serves a vital function. It automatically diverts blood away from poorly ventilated (low oxygen) areas of the lung and redirects it to areas that are well-ventilated (high oxygen). This process, illustrated in Figure 39-3, ensures that blood is sent where it can pick up the most oxygen, maximizing the efficiency of gas exchange.

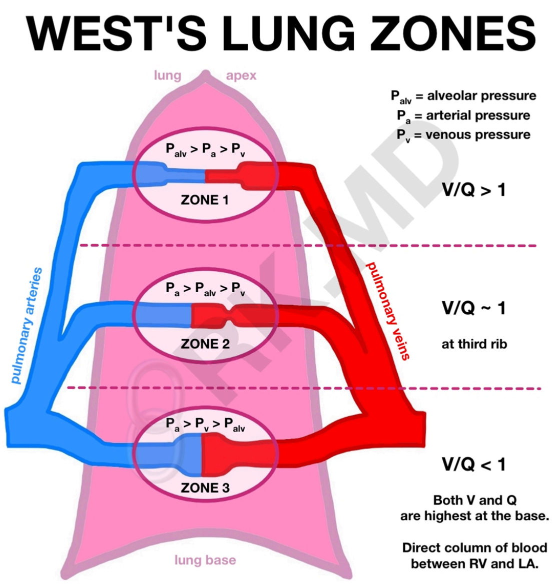
How Does it Work?
- The exact cellular mechanism is not fully understood, but it's believed that low oxygen levels may:
  - Cause the release of vasoconstrictor substances.
  - Inhibit the release of vasodilators like nitric oxide.
  - Block oxygen-sensitive potassium channels in the smooth muscle cells of the pulmonary arteries. This leads to an influx of calcium, causing the muscle to contract and the vessel to constrict.

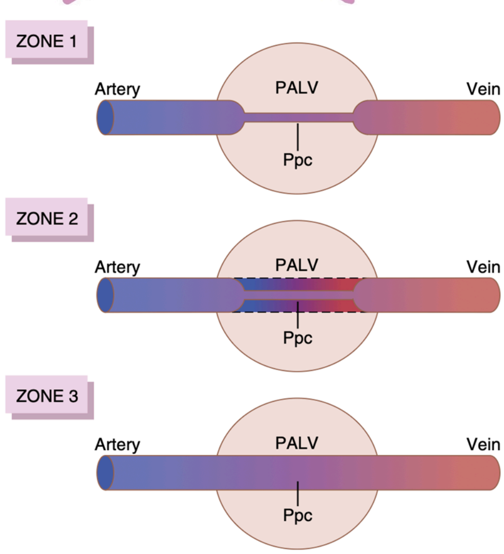
The Effect of Gravity (Hydrostatic Pressure)
- **Main Concept:** In an **upright person,** gravity creates a hydrostatic pressure gradient in the lungs. Blood pressure is lowest at the top (apex) of the lung and highest at the bottom (base).

- **Consequence:** This pressure difference causes a very uneven distribution of blood flow. As shown in Figure 39-4, at rest, there is significantly more blood flow through the bottom of the lungs than through the top.

The Three Zones of Pulmonary Blood Flow

- To explain this gravity-dependent flow, the lung is conceptually divided into three zones. The pattern of blood flow in each zone is determined by the relationship between alveolar air pressure (PAlveolar) and pulmonary arterial pressure (PArterial).
- Zone 1: No Flow
- Zone 2: Intermittent Flow
- Zone 3: Continuous Flow





Zone 1: No Flow 🛑

- **Condition: Alveolar air pressure is greater** than the arterial blood pressure at all times (PAlveolar>PArterial).
- **Result:** The capillaries are compressed shut, and there is no blood flow.
- **Key Point:** This is an abnormal condition and is not found in a healthy, resting person. It can occur with severe blood loss (very low blood pressure) or positive pressure ventilation (very high alveolar pressure).

Zone 2: Intermittent Flow 🚦

- **Condition: Arterial pressure is higher** than alveolar pressure only during the peak of the heart's contraction (systole). During relaxation (diastole), the arterial pressure drops below alveolar pressure (PArterial(systole)>PAlveolar>PArterial(diastole)).
- **Result**: Blood flows in pulses, only during systole.
- **Location:** This is the normal pattern in the apices (tops) of the lungs of an upright person at rest.

Zone 3: Continuous Flow ✅

- **Condition**: **Arterial pressure is always greater** than alveolar air pressure, throughout the entire cardiac cycle (PArterial>PAlveolar).
- **Result:** The capillaries remain open, and blood flows continuously.
- **Location:** This is the normal pattern in the lower and middle parts of the lungs in an upright person. When a person is lying down, the entire lung experiences Zone 3 blood flow.

Effect of Exercise

- During exercise, cardiac output and pulmonary arterial pressure increase significantly.
- This rise in pressure is enough to **convert the Zone 2** areas at the top of the lungs **into Zone 3 flow.**
- The result is a more even distribution of blood flow, with increased flow through all parts of the lung to maximize gas exchange.